

Visium HD – Analysis

Author: Naomie Pont

Workflow

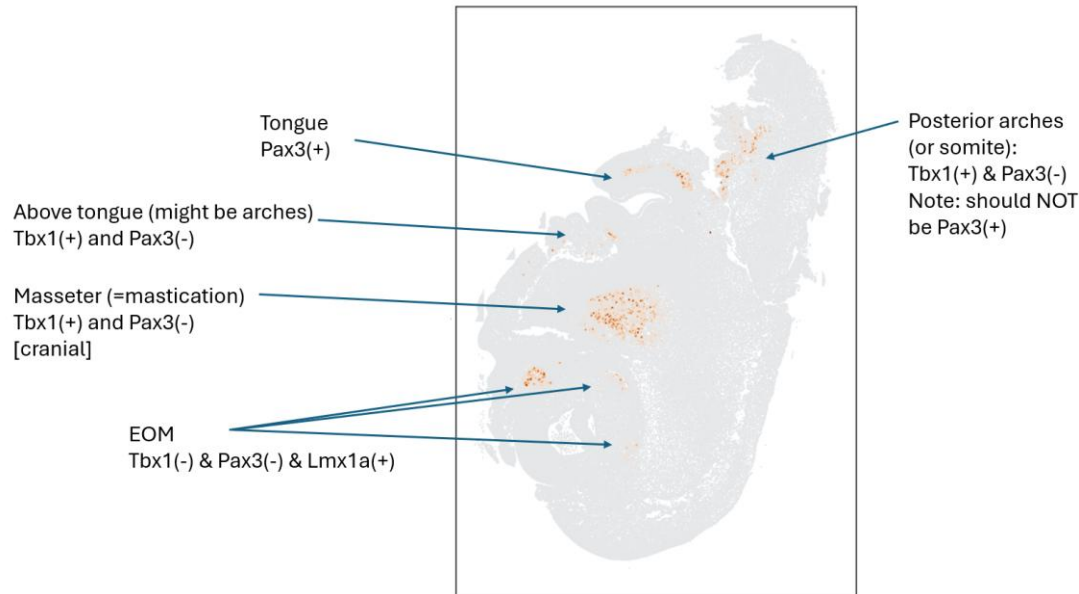
- Tissue segmentation with StarDist
- Definition of Anatomical clusters (=muscle clusters)
- Subset tissue for muscle clusters only
- Differential Expression Analysis:
 - o at single cell level (2195 cells x 2000 genes if keeping highly variable genes only)
 - o at pseudobulk level (72 samples × 642 genes after gene filtering)
- Transcription Factors Enrichment Analysis

I use scanpy which work on AnnData objects.

Definition of Anatomical Clusters

Stratify the tissue into 5 anatomical clusters as follows:

Figure1: expression of MyoG on slide D1, with known muscle signatures (from mouse).



Method:

1. Identify cells expressing Myogenic Regulatory Factors (MRFs): Myf5 and/or MyoG and/or MyoD1
2. Define 5 muscle groups based on spatial coordinates (Region of Interests – ROI - manually defined using Fiji software).
3. Use the following conditions to define 5 anatomical clusters:
 - a. EOM = { in EOM ROI } & {Pax3 negative} & [{ cells expressing at least one MRF } or {Tbx1 negative & Lmx1a positive & Pax3 negative}]
 - b. Masseter = { in masseter ROI } & {Pax3 negative} & [{ cells expressing at least one MRF } or {Tbx1 positive & Pax3 negative}]
 - c. Arches = { in arches ROI } & {Pax3 negative} & [{ cells expressing at least one MRF } or {Tbx1 positive & Pax3 negative}]
 - d. Above tongue = { in above tongue ROI } & {Pax3 negative} & [{ cells expressing at least one MRF } or {Tbx1 positive & Pax3 negative}]
 - e. Tongue = { in tongue ROI } & [{ cells expressing at least one MRF } or {Pax3 negative}]

Note1: we add the condition on Pax3 negative because the “or” condition might leak some Pax3 positive cells. The “or” condition is used because there can be cells that aren’t MyoG positive yet -those are stem cells- .

Note2: tongue is somehow the negative control as it is the only Pax3 positive group.

At this stage, we have 5 anatomical clusters on which to work.

Some numbers:

- 2.11 % of segmented cells express at least one of the Myogenic Regulatory Factors
- 39% of segmented cells are inside a muscle ROI
 - 386 cells in EOM.
 - 211 cells in tongue.
 - 363 cells in arches.
 - 95 cells above tongue.
 - 1140 cells in masseter.
 - 0.40 % of cells are EOM.
 - 0.22 % of cells are tongue.
 - 0.37 % of cells are arches.
 - 0.10 % of cells are above tongue.
 - 1.18 % of cells are masseter.
- We try the different combinations of factors to define EOM and thereby increase the number of cells:

[illegible]

Conclusion: the only combination of conditions that increases the number of cells is condition 8 but it corresponds to Pitx2 alone (not in combination with any MRF+), and therefore contains non-muscle cells. Therefore, we define EOM as condition 2 (although it leads to the same amount of cells as when not considering Pitx2+&MRF+ but it is more comprehensive/thorough).

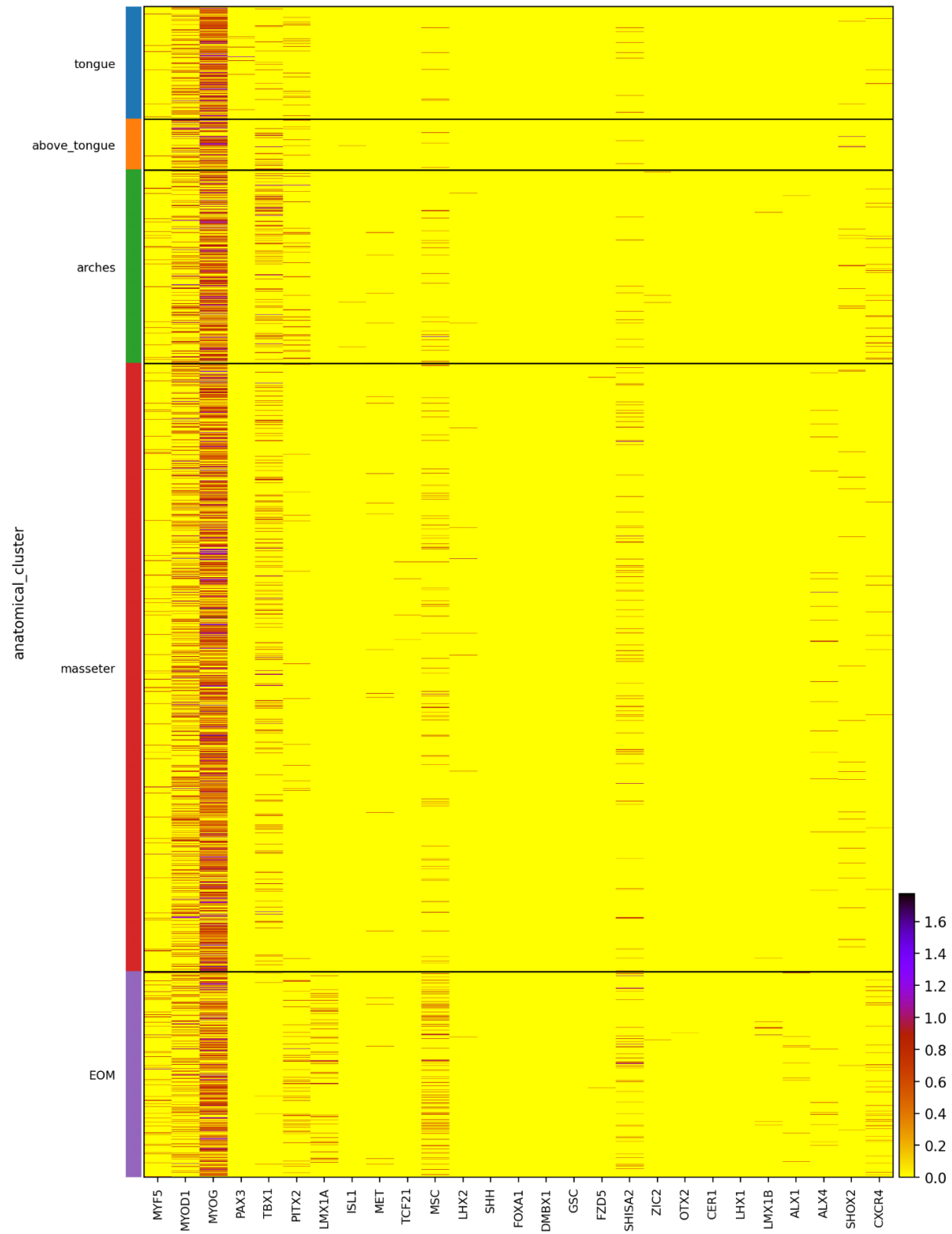


Figure2: Heatmap showing known markers expression (log-normalized). Single-cells are shown on each row.

Notes:

- there are 5 cells in EOM that express Tbx1. They come from condition “expressing at least one of the MRFs”. In other words: 5 cells in EOM express Tbx1 and at least one MRF
- I changed tongue to: (In tongue ROI) & [(Pax3+ & MRF-) or (Pax3- & any MRF+) or (Tbx1- & any MRF+)]
But it did not add cells to tongue being defined as: (In tongue ROI) & (Pax3+ or any MRF+)

Comments:

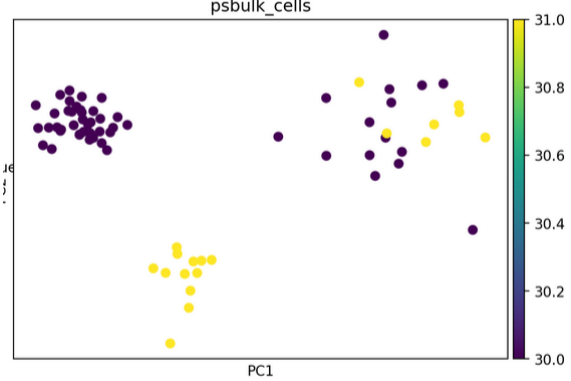
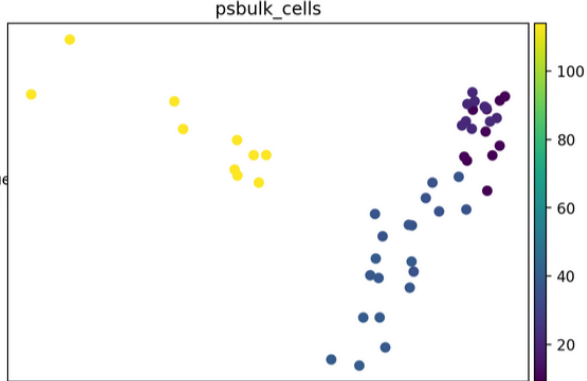
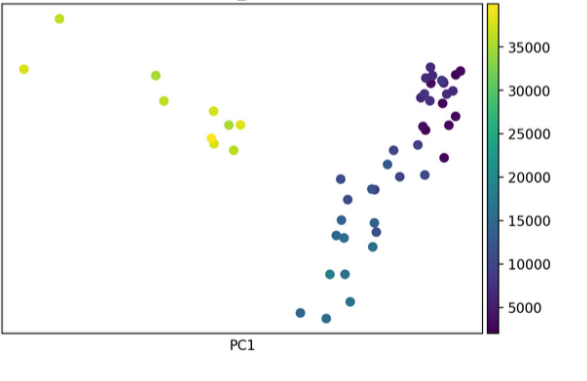
- About EOM:
 - o Isl1 not in EOM → good
 - o Lmx1a: only in EOM → good
 - o Plenty of Pitx2+ cells in EOM → good
 - o There are both cells co-expressing Lmx1a and Pitx2, or exclusively expressing one.
- About tongue:
 - o 11 (out of 211) tongue cells express Tbx1 (good or bad?)
 - o Only 7 (out of 211) tongue cells express Pax3 → bad. It means most of our tongue cells are cells that are in Region of Interest of tongue and that express at least one MRF (57 MYOD1 , 162 MYOG , 6 MYF5) (=differentiated cells?). Zooming in, we see that most tongue cells don't co-express the MRFs but rather exclusively express one of the MRFs.
- About arches:
 - o Isl1 only express here (and in “above tongue” which is a type of arches) → good. But really really poorly (in 3 cells) → bad
- General:
 - o RIPPLY, MRF4, SDF1 aren't in the expression data, can't be plotted on heatmap (while they were in my list of markers of interest)
 - o We struggle with sensitivity/depth.

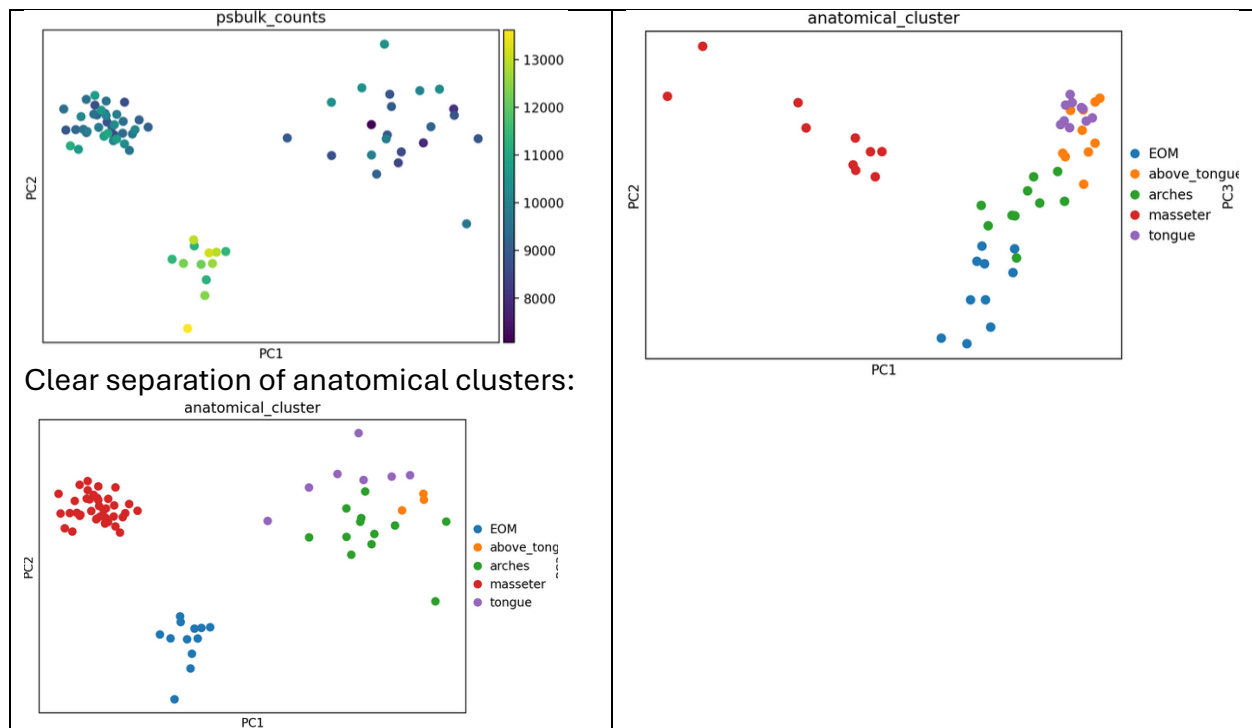
Pseudobulk Level

Differential Expression Analysis

Pseudobulks Generation

Method: create pseudobulks by aggregating cells owing to the same anatomical cluster (=muscle). Ensure pseudobulk samples are of same size (#cells) (rather than having the same number of pseudobulks per anatomical cluster) to reduce confounding factor. Indeed, when plotting the total counts in each pseudobulk (=psbulk_counts) and psbulk_cells on PCA space, we see these two drive variability and may confound the downstream analysis → rather define pseudobulks of same size to stabilize psbulk_counts. Note: still psbulk_counts drive a bit of variability but less than in the other method.

Pseudobulks of same size (→ different #psbulk in each cluster) [chosen method]	Same nb of psbulks in each cluster (→ different psbulks sizes) [not the chosen method]
 Looks like clusters are separated based on psbulk_cells but looking at the scale, it's the same value almost. Also we see separation of clusters based on psbulk_counts but the scale covers a lower range than in the other method, and there is not a gradient along one PC axis (like it's the case in the other method).	  Anatomical clusters are less well separated, and mostly separated based on psbulk_counts and psbulk_cells:



Clear separation of anatomical clusters:

This comes with the disadvantage of generating uneven group sizes (possibly impacting the DEA downstream, by inflating p-values especially in bigger groups). Eg way more pseudobulks in masseter than in “above tongue”.

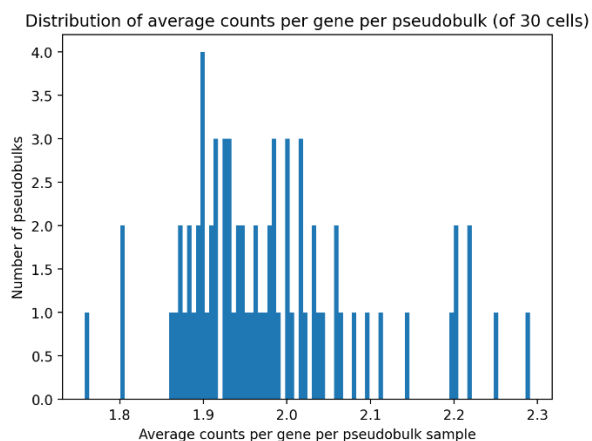
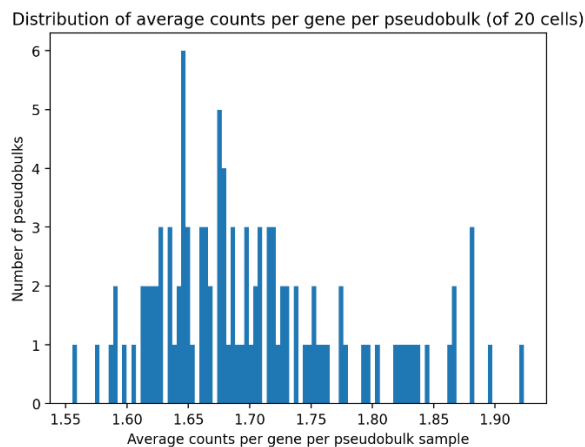
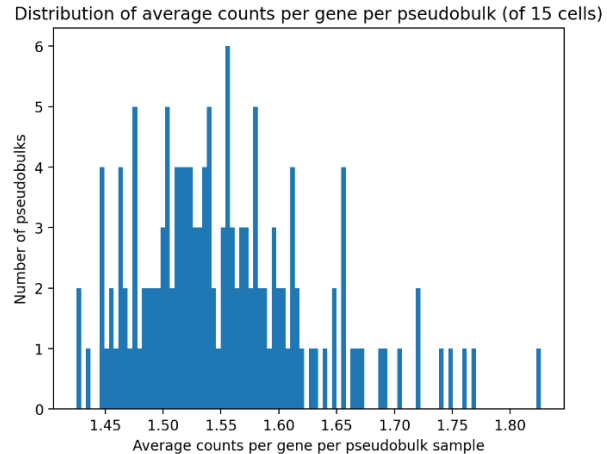
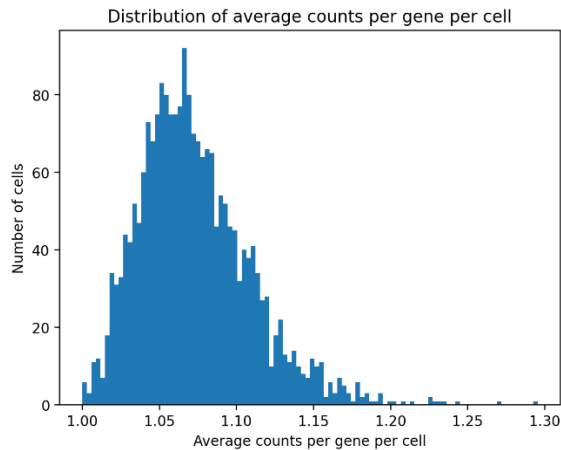
My algorithm for generating pseudobulks of same size:

- **Grouping:** cells are first grouped by Anatomical Cluster (muscle)
- **Filtering:** groups smaller than a minimum cell threshold (28) are excluded
- **Partitioning:** for each sufficiently large group, cells are randomly shuffled and divided into pseudobulks of target size (30)
- **Adjusting leftovers:** any leftover cells (if the group size isn’t perfectly divisible) are distributed one-by-one to some pseudobulks, so that their sizes differ by at most one cell
- **Labelling:** each cell is assigned a pseudobulk ID within its group
- **Filtering:** cells not assigned due to small group size are removed

Signal Optimization:

I compared the signal at the single cell level (=counts per gene per cell), and at pseudobulk levels (=counts per gene per pseudobulk) for increasing number of cells:

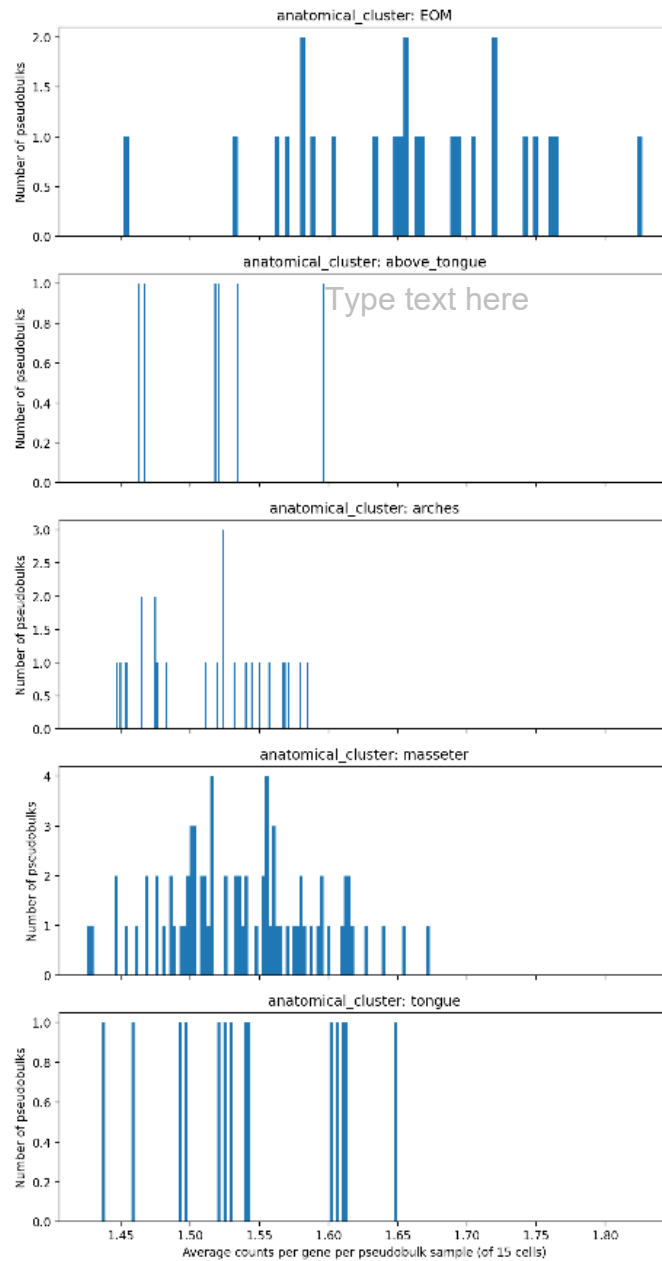
Goal: prove we gain from using pseudobulk. Especially because we deal with spatial transcriptomics which has lower capture efficiency compared to scRNA-seq.



➔ The number of counts per gene is doubled when considering pseudobulk and when aggregating more cells.

The signal should be fine with pseudobulk of at least 20 cells, especially knowing the data is very sparse: just 5K genes are expressed (>0 count) (and the count matrix contains 18K), so these averages are hiding a level of information: we're expecting some genes to have many counts while others have low counts, and the mean does not reflect this.

It's all about compromise: If we increase the nb of cells per pseudobulk too much, it generates unbalanced results (some anatomical clusters with lots of samples – like masseter - and others – like above_tongue – with very few) and forces us to filter our more cells (creating X samples with desired nb of cells and then not enough cells to fill in a new pseudobulk). Option: compromise.



[distribution per cluster, pseudobulks of

15 cells]

	Single cell level	Pseudobulk level
Advantage	- DEA muscle of interest versus. other muscles	- Noise reduction by aggregating counts - More robust (higher signal i.e. more counts per gene, less sparsity)

		- Does not consider cells as independent amongst a same sample
Drawback	- Noise/sparse because of dropout* and thus low signal (in terms of counts per gene per cell) - More comparisons, and considering cells as independent while they're not as they come from same sample → increases risk of false positive	- Pairwise DEA
	In both cases: we have unbalanced group sizes (EOM bigger for example than above tongue), the DEA will work and DESeq2 deals with uneven group sizes but it impacts the power.	

*dropout event = when a gene is observed at a low or moderate level in one cell but not detected in a cell of the same type. These dropout events occur due to the low amounts of mRNA in individual cells and inefficient mRNA capture, as well as the stochasticity of mRNA expression. Consequence: sparsity. The excessive zero counts cause the data to be zero-inflated, only capturing a small fraction of the transcriptome of each cell.

Pseudobulk:

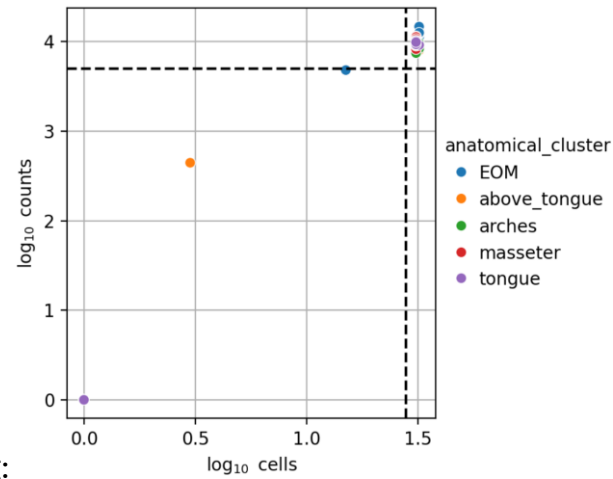
20 cells → uneven groups (=some anatomical clusters have more pseudobulks than others) but still compromise between increasing signal (=counts per gene per sample) and having sufficient nb of samples per group

30 cells → even better signal, but decreasing nb of samples in small groups (above tongue). Might be too poor in number of replicates.

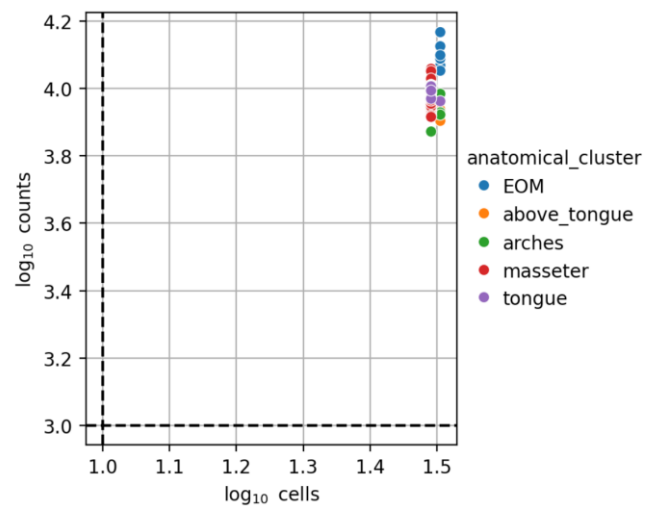
Our Final Pseudobulk of 30 cells:

We filter pseudobulks as explained in the algorithm above, based on a minimum number of cells per pseudobulk:

- set a minimum number of cells per sample such that they all have the same (keep just the bag at top right corner)
- set a minimum number of total counts per sample (y-axis)

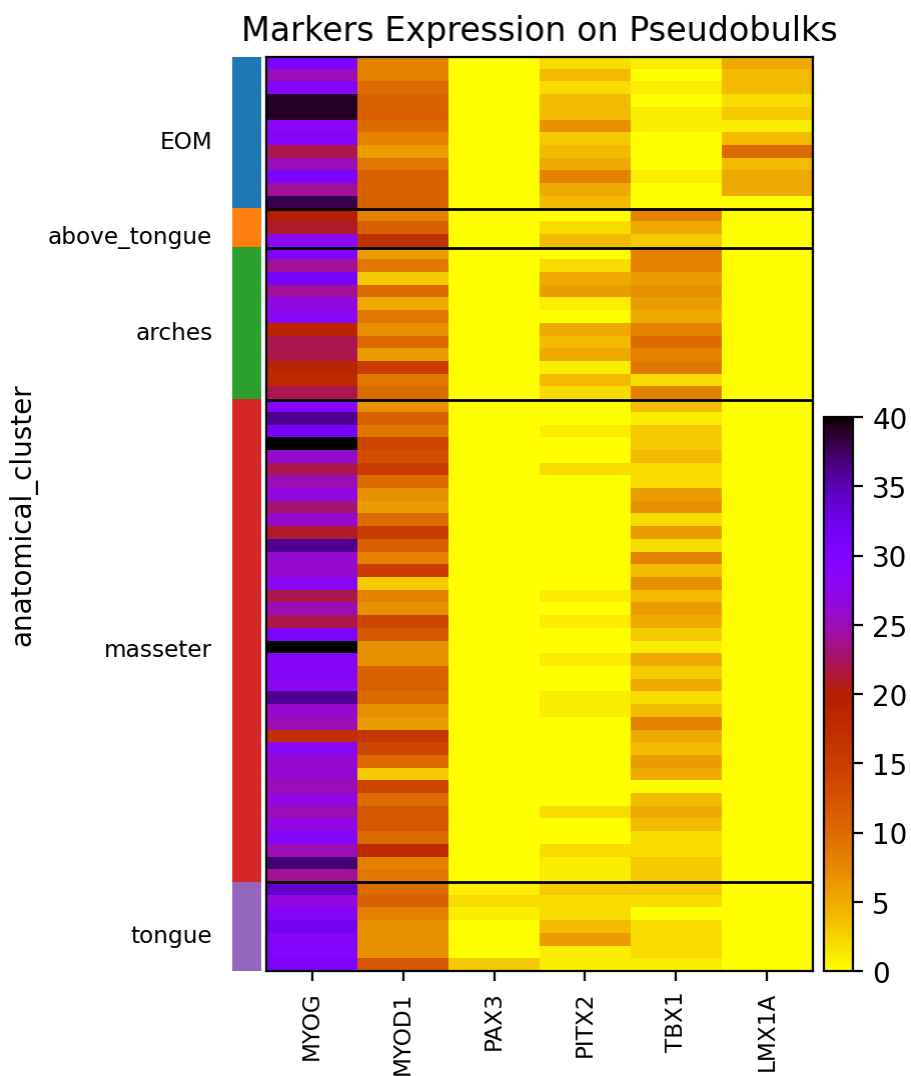


Before pseudobulks filtering:



After filtering:

Known markers used to define the anatomical clusters on my resulting pseudobulk:



	Number of pseudobulks of 30 (or 31) cells	Number of deleted cells because they're not enough to create a pseudobulk of 30 cells
Tongue	7 pseudobulks	16 cells
Above Tongue	3	2 cells
Arches	12	0
Masseter	38	0
EOM	12	14 cells

Filtering in pseudobulks:

Goal is to reduce noise, remove genes that are not properly expressed in the data, to not have too many genes for DEA. But then what is the balance between removing genes and not removing key genes? Also a risk of removing genes that are not expressed because of poor capture efficiency, while they're expressed in the tissue.

- *Filter by expression:*

Require a minimum counts in each pseudobulk for at least “smallest group size” (eg if using pseudobulks of 30 cells, above tongue group has 3 samples, so we keep genes with at least min_count counts in at least 3 samples) ; a minimum total number of counts across samples of each anatomical cluster ; a proportion of cells in the smallest group size that should express the gene

- *Filter by proportion:*

Require a gene to be expressed in a given proportion (“min_prop”) of cells for at least X samples (“min_smpls”). The data is very sparse because when using something bigger than 20% in at least 2 samples, we have very few genes.

Pseudobulk Size	Filtering Type	Filtering Parameters	Number of genes
25 cells	Filter by proportion	In at least 20% of cells of a sample for at least 2 samples	785genes
30 cells	Filter by proportion	In at least 20% of cells of a sample for at least 2 samples	642 genes

I try various filtering but trying to keep a sufficient number of genes for the DEA, especially as we have sparsity. Both lead to the same results. I'll show results of pseudobulk of 30 cells.

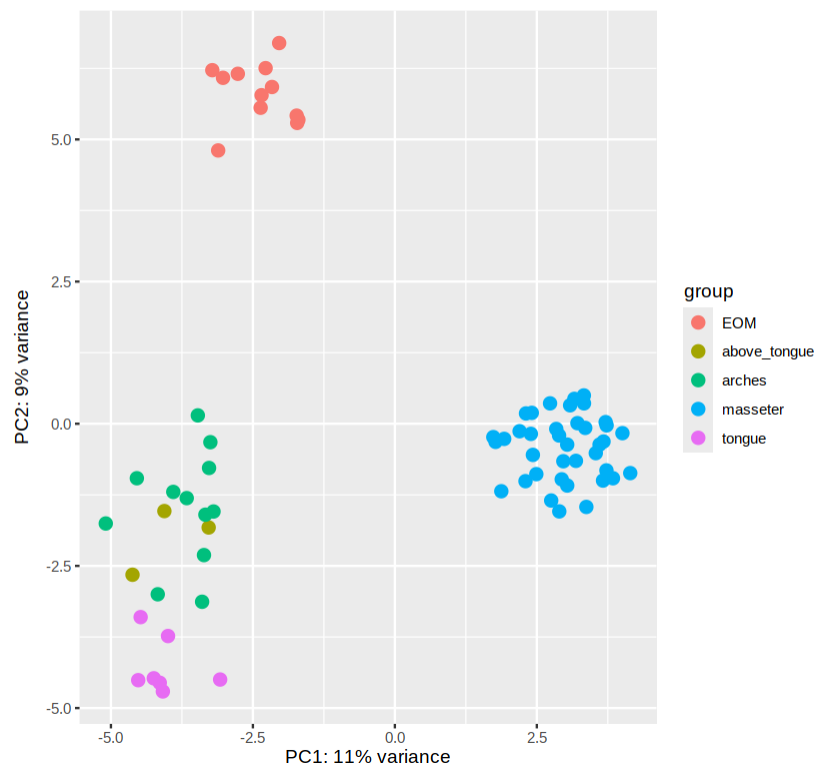
Why is multiple comparisons correction used?

- In DGE, you usually test **thousands of genes** to identify which ones are differentially expressed between conditions.
- Each gene is tested **independently** (e.g., using a statistical test like a t-test or Wald test), producing thousands of p-values.
- **Problem:** If you test **10,000 genes** with a significance threshold of 0.05, you expect about **500 false positives just by chance**.
- **Multiple comparisons correction** controls the **false positive rate** (Type I error rate) across all the genes tested.

Goal: find the compromise between removing too much signal (left with too few genes after genes pre-filtering) and having too many genes (increasing false positive rate)

Differential Expression Analysis (DEA) with DESeq2:

Based on Negative Binomial distribution and fits a GLM. First, check that there is no anatomical cluster with high within-group variability compared to the rest. If there is, exclude that group from the pairwise analysis. If no, fit the model on all and then use contrast argument in results to extract the comparisons of interest.



(pseudobulk 30 cells, 642

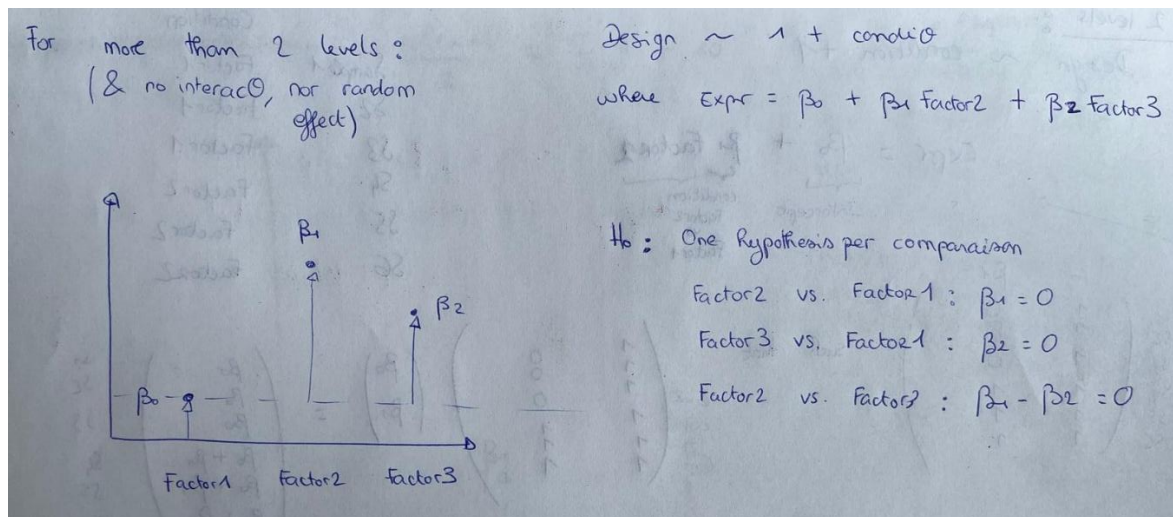
genes)

No group with bigger within-group variability. Apply design on all and then collect each comparison result (using contrasts).

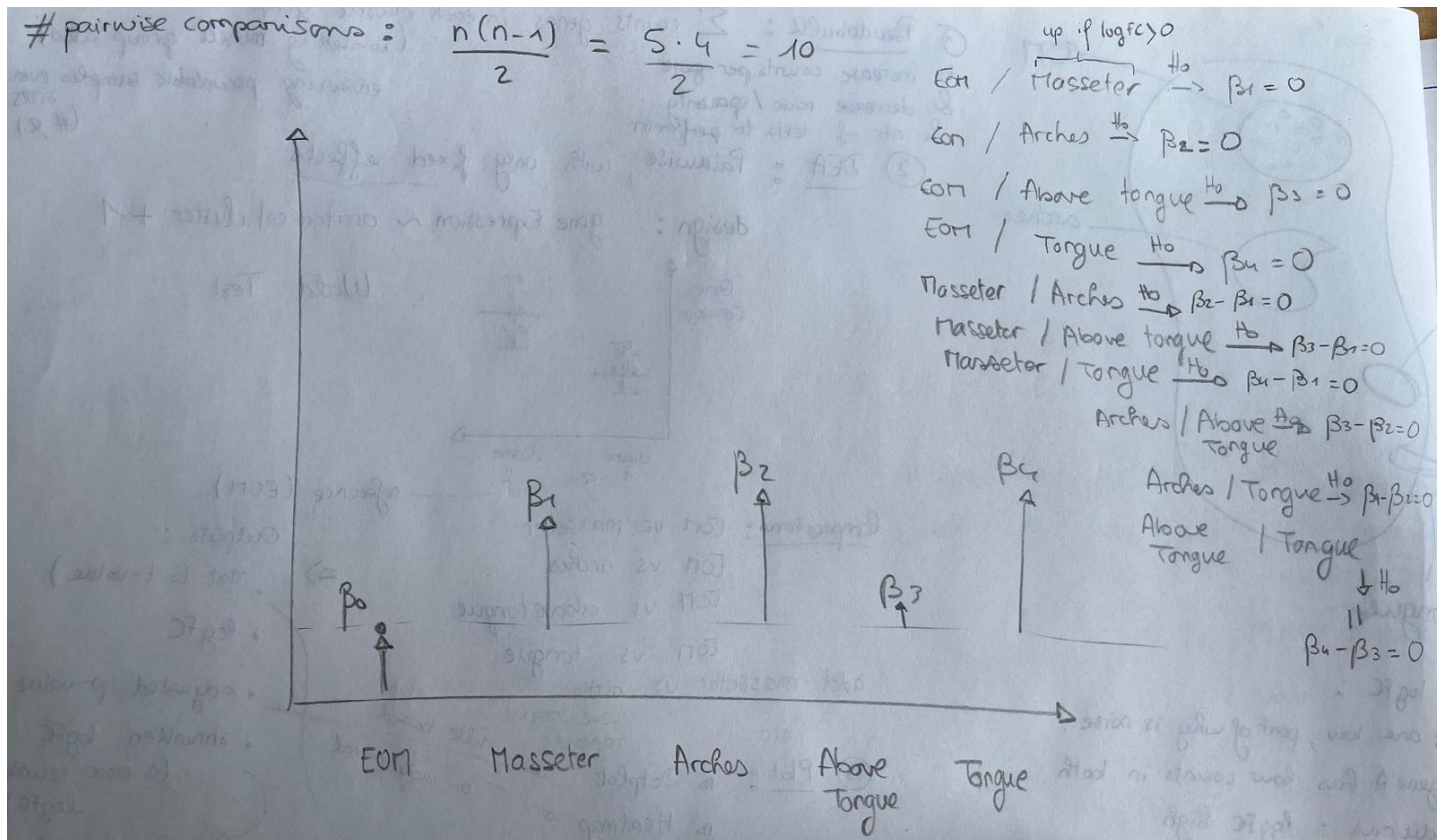
Design of DESeq2:

One factor with multiple levels. No interaction, only fixed effects. If we force the intercept to be 0 (design ~ 0 + condition, where condition is anatomical cluster) → we test a

coefficient corresponding to the average expression in each group. If we use intercept 1 (i.e. design ~ 1 + condition) → we test the *difference* between groups, i.e. design ~ 1 + condition).



[definitions]



[our situation]

LFCShrinkage:

Goal: When genes are noisy, the logFC aren't reliable. Imagine a gene has low counts in both conditions being compared, the logFC will be mistakenly high. We can use "shrinkage" which is a method that compares genes with low or high counts to re-adjust the logFC. It does not impact the p-values. I use these shrunken logFC values to check for the results and to select genes upregulated.

I use `lfcShrink()` function of DESeq2 and with `type="apeglm"`.

- ➔ Used also to check coherence of results: for some genes (e.g. the most significant ones), plot their normalized expression in each group (plotCounts function of DESeq2) and verify that the ratio between groups (computed manually by looking at plot) matches the shrunken logFC.

Types of comparisons in DEA:

there exist several which I describe here but I performed pairwise comparisons

- **One cluster vs. average of others:**

e.g. it tests for each gene: if $(\text{Average expression in EOM}) - \frac{1}{4}(\text{Average in masseter} + \text{Average in above tongue} + \text{Average in tongue} + \text{Average in arches}) = 0$

It's like testing if average expression of a gene in EOM is different from the expression averaged in all other four clusters.

Note that the average are actually in log scale, which is fine as we only want to test for the equation above, i.e. the significance, but not interested by the log FC

Design $\sim 0 + \text{condition}$

Contrasts = list (c(conditionEOM),

c('conditionmasseter','conditiontongue','conditionarches','conditionabove_tongue'),

listValues=c(1,-1/4))

Drawback: the four clusters to which EOM is compared aren't similar, so considering their average might be masking real effects and therefore not meaningful. And it's not testing the differences themselves but the averages.

- **Pairwise Comparisons:**

All anatomical clusters are compared, generating 10 comparisons.

- EOM vs. Arches
- EOM vs. Above Tongue
- EOM vs. Masseter
- EOM vs. Tongue
- Arches vs. Above Tongue
- Arches vs. Masseter
- Arches vs. Tongue
- Above Tongue vs. Masseter
- Above Tongue vs. Tongue
- Masseter vs. Tongue

Design $\sim 1 + \text{anatomical cluster}$

Advantages: it is the usual type of design. More control over what's being compared: truly compares the signal between two muscles; knowing we already have poor signal (see part "distribution of average counts per gene per sample") we need a good starting material, and grouping muscles might add a level of noise.

Drawback: are we really interested by the pairwise differences? or rather by the signal of one muscle vs. all other muscles? Additionally, it increases the number of tests and thus the False Positives rate.

DEA Results Processing

- Select Upregulated genes: for comparisons “group1_vs_group2”, genes upregulated in group1 correspond to shrunken $\log FC > 0$ while genes upregulated in group2 are genes with shrunken $\log FC < 0$. And we filter for significant results based on adjusted p-value < 0.05 (using Benjamini-Hochberg for multiple testing correction)
- Top upregulated genes: to increase confidence I select genes appearing in half of comparisons as significantly upregulated. Example: for all comparisons involving EOM (EOM vs. arches, EOM vs. above tongue, EOM vs. masseter, EOM vs. tongue) I select genes that are reported as upregulated in at least two of those comparisons.
- Plot results: dotplots and heatmap. On dotplots I label in bold gold the TF (not possible on the heatmaps because can't custom labels).

Transcription Factors Enrichments Analysis

Based on https://decoupler.readthedocs.io/en/latest/notebooks/scell/rna_psbk.html

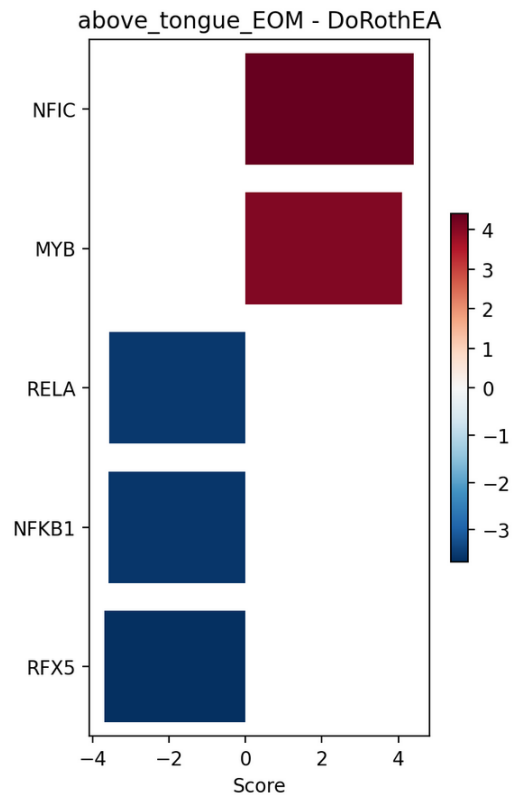
➔ Decoupler

Using [univariate linear model](#)

Using two databases of TF and its transcriptional targets: DoRothEA and CollecTRI. They don't fully overlap.

This TF enrichment analysis uses the DEA results as input: more specifically, the statistics of the Wald test, for all genes (not only the significant ones). Then I select significantly enriched TFs by filtering for adjusted p-values < 0.05 . A positive enrichment score (ES) corresponds to a TF enriched in group1. A negative ES is a TF enriched in group2. When we provided the results of each pairwise comparison group1_vs_group2 as input.

Example: in the following comparison, the positive scores correspond to TFs enriched in “above_tongue” and the negative scores correspond to TFs enriched in EOM.



Results

Html reports:

"fast_npz\Naomie\Shahragim_Project\visium_HD\spatial_EOM\segmentation\StarDist\Markers_Identification_Pseudobulk_30cells_642genes.html"

Single Cell Level

Important note:

I couldn't apply DESeq2 for differential expression analysis because many genes have mostly zero across cells. Indeed, DESeq2 estimates size factors, and this throws the error that many genes contain zero (even after I filter for genes expressing at least X counts in Y cells). The problem arises because size factors estimation relies on geometric means which involve log while $\log(0)$ is not defined.

According to [this conversation](#):

One solution is adding 1 to all counts, but while it would work as a workaround, it is unclear what the implications are for the downstream analysis. This phenomena means samples

are under-sequenced or there are any other kinds of dropout events. It is in any case not normal and should probably not be ignored by just adding a pseudocount. **The error can mean we're applying DESeq to a single cell dataset with bad cells.**

➔ I rather use the usual scanpy method for DEA.

➔ But additionally, message of scanpy:

⚠ Warning

Comparing between cells leads to highly inflated p-values, since cells are not independent observations [Squair *et al.*, 2021]. Especially in single-cell data, consider instead to use more appropriate methods such as combining pseudobulking with [PyDESeq2 documentation](#).

`decoupler.pp.pseudobulk()` or `scanpy.get.aggregate()` can be used to aggregate samples for pseudobulking. Ours is a bit more verbose, but supports [Dask](#) arrays for improved performance.

So be aware that we might have quite many False Positive upregulated genes.

Pre-processing

Number of cells in segmented: 96978

Number of cells after selecting only for our muscle cells (cf “Definition of Anatomical Clusters”): 2195

I filter genes to reduce dimension: pick the top 2000 most highly variable genes.

TF enrichment Analysis

Using also decoupler univariate linear model. With DoRothEA and CollecTRI databases.

- Then I select for TF reported as enriched with an adjusted p-value < 0.05 and ¼ of cells for each anatomical cluster.

I plot all enrichment scores of those TF in “Heatmap of Enrichment Scores for TFs being significantly enriched or depleted in a 1/4 of cells of each anatomical cluster”.

And I plot the same heatmap, showing TF enriched in more than ¼ of cells of each anatomical cluster, but only coloring the pairs for which this significance is found. It's in “Heatmap showing significant pairs {muscle, enriched TF} only, when significance is found in more than 1/4 of cells for the given muscle”.

Example: ETS2 is found to be enriched with an adjusted p-value < 0.05 in more than $\frac{1}{4}$ of cells of cluster “arches”. ESR1 is **not** found to be significantly enriched in at least $\frac{1}{4}$ of cells for any cluster apart in masseter.

Then I plot the heatmap showing not the enrichment score but rather the expression of those enriched TF. Most of them aren’t expressed in our data, but as expected as the analysis relies on the target genes. TF RNAs can quickly degrade while they’re active.

- Then in section of the html report called “*TF differentially expressed across muscles*”:

We test and report which features (TF Enrichment Scores) are differentially active across our muscle clusters. For each muscle I only show the top 30 TF.

It is similar to a differential expression analysis but it relies on TF enrichment scores rather than expression values.

Differential Expression Analysis

One cluster versus the rest

I use function `scanpy.tl.rank_genes_groups` to perform a t-test and select for significant genes (based on adjusted p-value < 0.05 where it was corrected for multiple comparisons using Bonferonni Hochberg).

The “rest” corresponds to the union of every cells which are not in the cluster.

Example: “EOM vs. rest” where rest is the union of cells from arches, masseter, tongue, above tongue. It might not be really meaningful as the rest contains very distinct cells. I would rather consider the pairwise comparisons.

Results: dotplot (TF labeled in bold gold) and heatmap.

Pairwise comparisons

Same as for one cluster vs. rest but considering each pair of muscle to perform a pairwise DEA. Using t-test, adjusted p-value < 0.05 .

Results: dotplot (TF labeled in bold gold) and heatmap.

In the file "Megabulk _ Genes from Single Cell Analysis.pdf" and in "Megabulk _ Genes from Pseudobulk Analysis.pdf":

I generated megabulks for each anatomical cluster (tongue, EOM, masster, above tongue, masseter) by aggregating the raw counts of all cells owing to an anatomical cluster.

I've read quite everywhere on the internet that it is not recommended to perform differential expression analysis when we have no replicate. I don't see how we could actually try as we can't estimate a within-group variance when we have one megabulk per condition (=anatomical cluster).

Therefore, I followed the recommendations of edgeR: "Be satisfied with a descriptive analysis, that might include an MDS plot and an analysis of fold changes. Do not attempt a significance analysis. This may be the best advice".

I've displayed the expression of the upregulated genes, derived from the pseudobulk (Megabulk _ Genes from Pseudobulk Analysis.pdf) and from the single cell analysis ("Megabulk _ Genes from Single Cell Analysis.pdf"). You can find for each heatmap, both a scaled version to appreciate the relative genes expression, and a not scaled version.

Note that I've used the raw counts and normalized the megabulks using log CPM. This is what is plotted on the heatmaps.